

CryptoTrace I4C

AUTOMATED BLOCKCHAIN FUND-FLOW TRACING & VASP IDENTIFICATION

for Indian Cybercrime Investigation

PROBLEM STATEMENT

SIH26183

ORGANISATION

Ministry of Home Affairs / I4C

CATEGORY

Software | Theme: Blockchain & Cybersecurity

DOCUMENT TYPE

Internal Team Reference — Fixed Master Report

CRYPTO
TRACE

FIXED MASTER REPORT — Complete reference from basic understanding → practical architecture → prototype → validation

This is an internal hackathon build document, not an official I4C document and not legal advice.

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§ 1

Executive Summary

In one sentence:

Given a victim-reported cryptocurrency wallet address, CryptoTrace I4C automatically retrieves public blockchain transactions, traces the fund flow for a limited number of hops, checks destination addresses against a curated VASP/exchange label database, shows confidence and risk indicators, and generates a structured investigation report.

What the Team Is Actually Solving

A cybercrime investigator may receive a wallet address from a victim. The address is public, but following money manually can involve many transactions, wallets, and browser tabs. CryptoTrace I4C automates the repetitive, evidence-gathering portion of that workflow. It does **not** identify the real person behind an exchange account, and it does **not** replace legal action.

Core Investigation Pipeline



Defensible Innovation

The strongest claim is not *'we used AI.'* Transaction retrieval, graph construction, and basic visualization already exist in public tools. The defensible opportunity is the **focused investigator workflow**: one suspect address → automated trace → nearest supported VASP attribution → evidence view → structured report, designed around the stated Indian cybercrime use case.

■ **Hard Boundary — Do NOT claim:**

Real-person identification • Guaranteed attribution of unknown wallets • Complete cross-chain tracing • Mixer de-anonymization • Automatic freezing of funds

§ 2

Original SIH Problem Statement — Verbatim

Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Background: Cyber fraud victims increasingly report suspect cryptocurrency wallet addresses used by fraudsters for collection of funds in cases involving investment scams, task-based frauds, sextortion, ransomware, phishing, darknet transactions, and organised cyber-enabled financial crimes.

During investigations, the reported wallet addresses are often non-custodial wallets, temporary burner wallets, or intermediary wallets used for layering and laundering. The inability to quickly identify the cryptocurrency exchange or VASP associated with these wallets delays freezing of assets, preservation of evidence, tracing of fund flows, and victim fund recovery.

Manual blockchain tracing requires significant technical expertise and time, particularly in cases involving multi-chain transfers, DeFi protocols, mixers/tumblers, bridges, and privacy-enhancing mechanisms.

The proposed solution envisages a **Real-Time Crypto Fraud Attribution System** capable of automatically analysing victim-reported wallet addresses and identifying the nearest exchange or VASP receiving direct deposits.

The system should:

- Ingest wallet addresses reported through cybercrime complaint systems
- Automatically perform blockchain tracing
- Identify associated exchanges or VASPs
- Detect fund movement patterns
- Generate actionable intelligence for investigators

Key features may include:

- Blockchain transaction graph analysis
- Clustering of exchange wallets
- Detection of intermediary laundering wallets
- Identification of cross-chain fund movement
- Integration with SAHYOG and NCRP platforms
- Automated alert generation
- Risk categorisation of wallets

Expected Solution: A software platform capable of real-time blockchain intelligence generation, automated VASP identification, tracing of suspect wallets, cross-chain transaction analytics, fund-flow visualisation, integration with LEA systems, and generation of standardised investigation reports.

The system should reduce response time in cyber fraud investigations, improve freezing of proceeds of crime, enhance coordination with VASPs, and strengthen digital evidence collection capabilities. The platform should further support API integrations, scalable blockchain indexing, AI/ML-assisted risk detection, and automated pattern recognition for fraud typologies.

Organisation	Ministry of Home Affairs
Department	Indian Cyber Crime Coordination Centre (I4C), CIS Division
Category	Software

Theme	Blockchain & Cybersecurity
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§ 3

What I4C Is Asking For — In Simple Language

Situation: A victim says: 'I lost money and the fraudster gave me this wallet address.' The investigator needs to discover where that money moved next and, ideally, which exchange/VASP ultimately received it.

What the System Should Do

1	Accept the victim-reported suspect wallet address.
2	Detect or accept the blockchain network.
3	Retrieve public transaction data automatically.
4	Follow relevant outgoing fund flows across a controlled number of hops.
5	Identify known VASP/exchange destinations when evidence supports the attribution.
6	Highlight intermediary/risk patterns without treating heuristics as proof.
7	Present the fund path visually and in plain language.
8	Generate structured intelligence/evidence for the investigator's next legal/compliance step.

Three Layers You Must Never Confuse

Layer	What it means	Can CryptoTrace do it?
Blockchain data	Address A sent X asset to Address B at time T; transaction hash and public fields.	■ YES
Entity attribution	The destination address is associated with a known exchange/VASP.	■ YES — when label evidence exists; confidence must be shown.
Real-world identity	The exchange account is registered to a specific person.	■ NO — requires off-chain/legal process with the VASP.

§ 4

Blockchain & Crypto Fundamentals

Reference for any team member unfamiliar with crypto terminology.

Term	Simple Meaning
Wallet address	A public-looking identifier used to receive/send crypto. More like an account ID than a person's name.
Transaction hash	Unique identifier for a blockchain transaction; useful as evidence for a specific transfer.
Blockchain explorer	Public interface/API for reading blockchain activity, e.g. Etherscan (ETH), TronScan (TRON).
VASP	Virtual Asset Service Provider. In this project, think mainly of cryptocurrency exchanges and regulated services.
Fund-flow tracing	Following money from the reported wallet through subsequent transactions.
Intermediary wallet	A wallet between the reported wallet and a later destination, often used to move/consolidate funds.
Burner wallet	A temporary wallet used for a limited purpose and then abandoned; an indicator, not proof of fraud.
Layering	Moving funds through multiple steps/wallets to make tracing harder.
Mixer / tumbler	A service designed to make fund provenance harder to follow. A student MVP should detect/flag — not defeat.
Bridge	A mechanism for moving assets between blockchain ecosystems. Cross-chain tracing is much harder.
DeFi	Financial applications implemented through smart contracts. Can complicate interpretation of fund flows.
KYC	Know Your Customer information held by a VASP. Public blockchain data does NOT contain this identity information.

What Public Blockchain Data Can and Cannot Show

■ CAN Show	■ CANNOT Show
• Transaction hash, timestamp, sender, receiver, asset/token, and amount • Transaction history for a public address • Interactions with known smart contracts and observable patterns • Known entity label when a trusted source maps the address to an exchange	• The real person's name, phone number, or government ID • Physical location • Guaranteed common ownership of two wallets • The customer identity of an exchange account

§ 5

How Crypto Fraud Happens in Practice

Educational model — do not present fictional examples as verified cases.

Common Simplified Pattern



S t a g e	What Happens	Where Blockchain Appears
1	Scammer builds trust or offers a fake investment/task opportunity.	Usually off-chain communication.
2	Victim is shown fake profits or pressured to pay more.	Mostly off-chain.
3	Victim buys a crypto asset such as USDT.	On-chain purchase/withdrawal may occur.
4	Victim sends crypto to a wallet supplied by the fraudster.	Public blockchain transfer.
5	Funds may move through intermediary/consolidation wallets.	Public blockchain transfers.
6	Funds may reach a VASP/exchange or another service.	Public transfer may be observable; account identity is not.
7	VASP may connect deposit to a customer account via internal records.	Off-chain; legal/compliance process.

Fraud Patterns Relevant to the Problem Statement

Investment / trading scams: Victims persuaded to send money into a fake investment flow; crypto becomes the transfer rail.

Task-based scams: Victims receive small initial rewards and are later asked to deposit more money; crypto may be used.

Sextortion: Victim is threatened and asked to pay crypto; the on-chain path may be comparatively short.

Ransomware: Cryptocurrency demanded as ransom; the surrounding cyber incident differs from ordinary financial fraud.

Layered laundering: Funds move through several wallets, consolidation addresses, bridges or other services.

Demo rule: Use a publicly documented case with a trace you can independently verify. If a fictional victim/case is used for UI, label it clearly as a demo scenario. Never invent a real complaint ID or imply government access you do not have.

§ 6

Current Investigation Workflow & Pain Points

Simplified Manual Workflow

- 1 Complaint is received with a suspect wallet address and other case details.
- 2 Investigator determines which blockchain is relevant.
- 3 Investigator opens a blockchain explorer and inspects transactions.
- 4 Investigator follows outgoing transfers from the reported wallet.
- 5 The process repeats for additional hops and branches.
- 6 Investigator checks whether destination addresses correspond to known exchanges/services.
- 7 Findings and transaction hashes are manually documented.
- 8 The appropriate legal/compliance process is used to contact the VASP.

Where CryptoTrace Automation Fits

Manual Work (Investigator)	CryptoTrace Automation
Identify chain from address/input	Chain validation / selection
Open explorers repeatedly	API-based transaction retrieval
Follow wallet-to-wallet transfers	Graph construction + bounded N-hop traversal
Check exchange addresses	Curated VASP label matching
Spot suspicious relay/consolidation patterns	Transparent deterministic heuristics
Write repetitive findings	Structured evidence/report generation
Legal action / obtain KYC	■ Human investigator + formal process — outside prototype

Note: Earlier reports included hard numbers such as '4–6 hours per case' or '38,000 complaints/month.' Unless the team has a current primary source in hand, treat such numbers as claims to verify, not guaranteed facts. For the prototype, measure your own before/after workflow time.

§ 7

Existing Solutions — What Is Already Solved?

The team should not claim to have invented blockchain analytics itself.

Category	Examples	What They Provide	Implication for Us
Public explorers / APIs	Etherscan, TronScan, BscScan	Transaction lookup and public blockchain data	Raw data access is not our innovation.
Commercial investigation platforms	Chainalysis, TRM Labs, Elliptic, Crystal	Graph analytics, entity attribution, risk/compliance workflows, large datasets	Do not compete on database depth or chain count.
Free/open-source analytics	GraphSense, Breadcrumbs	Graph analysis / investigation building blocks	Visualisation alone is not differentiation.
Community/entity labels	Explorer tags, public label repos	Known addresses associated with exchanges/services	Build a smaller, transparent label layer for the MVP.

What This Means for Our Project

Already solved:

- Fetching public transaction data
- Basic graph visualisation
- Known-address matching
- Many forms of professional blockchain analytics

Partially solved:

- End-to-end workflow for small teams
- India-specific investigator UX
- Curated local context
- Transparent report output

Our opportunity:

- A narrow, reproducible workflow that is easier to use and demonstrable within hackathon constraints

Positioning: Do NOT say 'No one has built blockchain tracing before.' That is false. Say: 'Existing tools are powerful; our prototype focuses on a narrow, transparent investigator workflow for the stated use case.'

§ 8

Real Gap & Innovation Test

Use this chain to justify every feature: *Existing capability* → *Limitation for target workflow* → *Specific gap* → *Our focused addition* → *Measurable demo*.

Existing Capability	Limitation / Gap	CryptoTrace Opportunity
Explorer shows transaction data	Investigator still performs repeated manual hops and documentation	Automate bounded tracing from one reported address.
Graph tools visualise networks	Generic graph does not automatically produce the investigator's VASP-oriented workflow	Highlight path to supported VASP and evidence.
Commercial tools have large entity databases	High-end data/enterprise workflows are outside a student MVP	Use a small, transparent, source-labelled VASP dataset.
Heuristics can detect patterns	Heuristics can produce false positives	Expose rule + confidence; never treat heuristic output as proof.
Reports can be generated	Generic reports may not follow the investigator workflow	Generate a structured case/evidence summary tied to transaction hashes.

Genuine Differentiation Statement:

'Unlike a generic blockchain explorer, CryptoTrace I4C turns a reported wallet address into a bounded, evidence-linked fund-flow investigation: trace → identify supported VASP → explain confidence → generate a structured report.'

Why This Is Stronger Than 'We Use AI'

The core evidence path is **deterministic**. AI, if used, should only convert verified structured results into readable language. That makes the system easier to test, explain, and defend before judges and investigators.

§ 9

Proposed Solution — CryptoTrace I4C

Primary user: Cybercrime investigators / analysts who need fast, understandable blockchain intelligence without manually navigating many transactions.

Input	Processing	Output
• Suspect wallet address • - Optional chain override • - Optional complaint/case ID • - Optional amount / depth • - Demo metadata	• Chain detection • - Public API transaction retrieval • - Transaction normalisation • - N-hop graph traversal • - VASP label matching	• Interactive transaction graph • - Fund-flow path + hop count • - Known VASP match + confidence • - Risk/heuristic indicators • - Structured PDF investigation summary

Recommended MVP Scope

Primary chains: Ethereum + TRON. TRON is especially relevant to USDT-TRC20 scenarios.

Optional: BNB Smart Chain only after the core path is stable.

Depth: 2–3 hops for a predictable prototype.

VASP database: A small, source-labelled set of known addresses rather than pretending to have a commercial-scale entity database.

Output: Graph + path + confidence + evidence hashes + PDF report.

Example system outcome: Reported wallet → intermediary wallet → known VASP address. The system displays the transaction hashes supporting the path, identifies the VASP only to the level supported by the label source, and tells the investigator what further legal/compliance step is required.

§ 10

Requirement-to-Feature Mapping

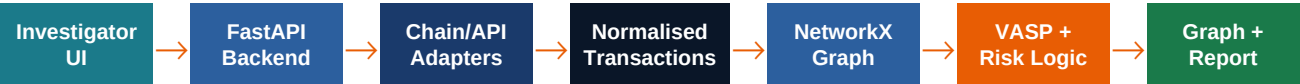
PS Requirement	Prototype Feature	How It Is Demonstrated
Real-time identification	Automated address submission → analysis pipeline	Show elapsed time on UI; use cached data for demo reliability.
Fraud-linked exchanges	Known VASP label database + transparent heuristics	Known destination becomes a labelled node; heuristic result shows rule and confidence.
Victim-reported suspect wallet	Single investigator input form	Paste the wallet address and optional case metadata.
Automated blockchain analytics	API retrieval + normalisation + graph traversal	No manual explorer hopping during the demo.
Nearest exchange/VASP receiving direct deposits	Bounded path search to supported labelled VASP nodes	Highlight path and show hop count/evidence.
Fund-flow tracing	N-hop traversal	Show wallet-to-wallet edges with amount/time/hash.
Actionable intelligence	Evidence panel + structured report	Report tells investigator what was observed and what requires legal follow-up.
Dashboards / visualisation	Interactive graph	Click nodes for address, labels, transaction/risk details.

§ 11

Technical Architecture

Build the smallest architecture that proves the problem statement.

End-to-End Pipeline



Module-by-Module Breakdown

Module	Responsibility	Implementation
Input / validation	Validate address and optional chain/depth	Frontend validation + backend validation
Chain detection	Determine likely chain from address format; allow manual override	Python utility
Transaction fetcher	Retrieve public transactions and token transfers	Etherscan / TronScan / BscScan APIs
Normaliser	Convert different API responses to one internal schema	Python dictionaries / Pydantic models
Graph builder	Create wallet nodes and transaction edges	NetworkX
Traversal	Explore outgoing flows to a bounded depth	BFS/DFS; shortest path where appropriate
VASP matcher	Match exact known addresses	JSON/SQLite label store with source + date
Risk heuristics	Flag observable patterns	Deterministic rules with explanations
Report generator	Create evidence summary	ReportLab
Dashboard	Visualise graph and findings	React + Cytoscape.js
Optional AI	Narrate verified results only	LLM receives structured JSON — never invents evidence

§ 12

VASP Identification Strategy

This is the most important part to explain honestly.

Method	How It Works	Confidence	MVP Use
Exact known-address match	Destination address exactly matches a labelled VASP address in your dataset.	High (assuming the source is trustworthy and current).	PRIMARY method.
Explorer/community tag	A public explorer or trusted public dataset labels the address.	High/medium depending on source quality.	Use as supporting evidence; record source.
Behavioural heuristic	Address shows exchange-like patterns: many deposits + consolidated withdrawals.	Medium/low; not proof of ownership.	Secondary investigative lead ONLY.
Commercial/proprietary entity intelligence	Large providers use proprietary labels, research and partnerships.	Potentially strong but not reproducible by student MVP.	Do NOT pretend to replicate it.

Suggested Label Record Schema

```
{ "address": "...", "chain": "TRON", "entity": "Example VASP", "type": "exchange", "source": "public label", "source_date": "YYYY-MM-DD", "confidence": "HIGH" }
```

Critical Design Rule: Every attribution must expose WHY the system believes it. 'Exact address match to label source X' is far more defensible than 'AI detected Binance.'

§ 13

Data, APIs & Evidence

Source	Use	MVP Role	Important Caution
Etherscan API	Ethereum transactions / token transfers	Primary ETH source	API terms/limits can change; verify current documentation before final demo.
TronScan API	TRON / TRC-20 activity	Primary TRON source	Verify current endpoints, access and rate limits.
BscScan API	BNB Smart Chain	Optional	Add only after core ETH/TRON workflow is stable.
Public address-label datasets	Known exchange/VASP addresses	Attribution seed data	Record source/date; labels can become stale or incomplete.
Publicly documented case reports	Validation / demo ground truth	Test cases	Use only cases whose path/outcome can be independently checked.

Data You Do NOT Need for the MVP

- Private NCRP/I4C complaint data
- Private exchange KYC records
- Government-only APIs
- Personal information scraped from victims or suspects

Synthetic data rule: Synthetic data is fine for demonstrating UI/graph behaviour when public ground truth is unavailable, but label it clearly. For a strong validation claim, prefer a public case with independently verifiable transaction evidence.

§ 14

Internal Transaction Data Model

Normalise every blockchain API response into one common structure. This prevents the frontend and graph logic from becoming chain-specific.

Field	Example	Why It Matters
chain	TRON	Identifies network.
tx_hash	abc123...	Evidence reference.
from	T...	Source node.
to	T...	Destination node.
asset	USDT	Asset being traced.
amount	500	Value transferred.
timestamp	2026-08-27T12:00:00Z	Ordering and timing.
block	12345678	Blockchain evidence context.
source_url	Explorer URL	Human-verifiable evidence link.

Once all chains use the same schema, NetworkX can build one graph model and the report generator can use the same evidence structure. Only the API adapters need to know chain-specific response formats.

§ 15

Risk & Heuristic Analysis

Use rules that are explainable and measurable. Do NOT call them 'AI fraud detection' unless you actually train and validate a model.

Rule / Signal	Possible Interpretation	System Output
Large proportion of received funds forwarded quickly	Possible relay/intermediary behaviour	Flag: rapid forwarding
Very little history before first observed receipt	Possible burner-like behaviour	Flag: limited prior activity
Many incoming addresses consolidated into one outflow	Possible consolidation	Flag: consolidation pattern
Known fraud seed address match	Known association from the seed source	Flag: source-linked high-risk lead
Known mixer interaction	Tracing may become unreliable	Flag: mixer interaction / stop or branch
Bridge interaction	Fund flow may continue on another chain	Flag: cross-chain boundary

False-positive policy: Every flag must show the rule that fired and its confidence/limitations. The system produces investigative leads, not automatic accusations.

§ 16

Where AI Should — and Should Not — Be Used

AI is optional. The project does not need an LLM to perform the core blockchain analysis.

Task	Use AI?	Reason
Fetch transaction data	■ NO	API call is deterministic.
Build graph	■ NO	Graph construction is deterministic.
Find shortest/bounded path	■ NO	Use standard graph algorithms.
Exact VASP address match	■ NO	Database lookup is more reliable.
Risk rule evaluation	■ NO	Rules should be reproducible.
Plain-English summary	■ OPTIONAL	LLM can turn verified JSON into readable language.
Investigator Q&A; over verified case data	■ OPTIONAL	Useful after core workflow is stable; must remain grounded in evidence.

Safe LLM Contract:

Give the LLM only verified structured results. It may explain them, but it must NOT invent addresses, transaction hashes, entities, amounts, identity claims, or analysis of data it did not receive.

§ 17

Technology Stack

Layer	Recommended Technology	Why
Frontend	React.js	Fast component-based UI; suitable for an investigator dashboard.
Graph visualisation	Cytoscape.js	Interactive network visualisation with node/edge interaction.
Styling	Tailwind CSS or simple CSS	Fast UI iteration; avoid spending days on design systems.
Backend	Python + FastAPI	Fast API development; easy integration with Python graph/data libraries.
Graph analysis	NetworkX	BFS/DFS/shortest-path and graph manipulation.
Blockchain APIs	Etherscan + TronScan; BscScan optional	Public transaction data for selected chains.
Entity store	JSON for MVP; SQLite if needed	Small, transparent and easy to ship.
PDF generation	ReportLab	Programmatic PDF generation without a browser.
Optional AI	Claude/Gemini API	Narration only after deterministic pipeline is stable.
Deployment	Local demo first; hosted only if stable	Avoid deployment risk during a short hackathon.

Scope discipline: Do NOT add technologies just to make the architecture look advanced. A smaller working system beats a large architecture that cannot be demonstrated.

§ 183–5 Day Prototype Build Plan			
Time	Backend / Data	Frontend / Demo	Exit Condition
Pre-day	API keys; test one known address; prepare label seed; choose demo case	Create React shell + graph component	One API response successfully obtained.
Day 1	Transaction fetcher + normalisation + basic NetworkX graph	Input form + graph rendering	Address → transactions → graph works.
Day 2	N-hop traversal + VASP exact-match database	Node colours + details panel + path highlight	Known VASP node reached and shown.
Day 3	TRON support + deterministic heuristics + evidence links	Risk panel + summary card + loading/error states	Full core investigation flow works on test case.
Day 4	PDF report + caching + integration	UI polish + report button	End-to-end demo works repeatedly.
Day 5	Bug fixes, backups, timing measurements	Rehearsal + judge Q&A; prep	No new features; stable demo.

■ MUST BUILD	■ NICE TO HAVE	■ DO NOT BUILD (MVP)
• Address input + validation • Transaction retrieval (ETH + TRON) • Normalised transaction model • 2-3 hop graph traversal • Interactive graph • Known VASP matching (source/confidence) • Evidence panel + PDF report • One reliable public demo case	• INR conversion • BSC support • Batch address input • More sophisticated heuristics • Optional grounded LLM summary	• Full cross-chain tracing (arbitrary bridges) • Mixer de-anonymisation • Real-world identity attribution • Live monitoring infrastructure • Real NCRP/I4C integration • 20+ chain support • Mobile app • Commercial-scale entity intelligence

§ 19

Prototype & Demo Design

The judge should understand the value within the first 60–90 seconds.

Recommended Screen Flow



Step 1 — Input: Investigator enters a publicly documented suspect wallet address.

Step 2 — Analysis: System detects/selects the chain and retrieves cached/live public data.

Step 3 — Graph: Nodes and transaction edges appear; the suspect node is visually distinct.

Step 4 — Trace: System follows a bounded number of hops; highlights route to a supported VASP.

Step 5 — Explain: Destination label, source, and confidence are shown. Risk flags show underlying rules.

Step 6 — Evidence: Transaction hashes and explorer references are visible.

Step 7 — Report: One click generates a structured PDF containing the case summary and evidence.

Demo Reliability Rules

- Pre-cache the chosen demo case so the presentation does not fail due to external API unavailability or rate limiting.
- Keep one backup case and screenshots/video backup.
- Run the complete demo repeatedly before the evaluation.
- Do not invent performance numbers — measure your actual system.

§ 20

Validation & Metrics

Use metrics that you can actually measure.

Metric	How to Measure	Good Prototype Evidence
Transaction retrieval time	Timestamp request start/end	Report median/observed time for your test case.
Graph generation time	Timestamp normalised data → rendered graph	Report observed time and graph size.
VASP exact-match correctness	Use publicly documented cases with known destination labels	State number tested, exact matches, and coverage.
Path reconstruction	Compare system path with independently documented path	Show matching transaction hashes/addresses.
Manual steps reduced	Perform same trace manually vs system	Show a before/after count or time.
Report generation time	Timestamp report creation	Observed generation time.
Label coverage	Known labelled addresses / relevant addresses encountered	State coverage honestly.

■ **Critical: Never claim '90% accuracy', '100% detection', 'under 60 seconds', or similar figures unless you have actually measured them under a defined test set. Targets are NOT results.**

§ 21

Limitations, Failure Cases & Honest Responses

Failure Case	What Happens	What Prototype Does	Judge Response
Unknown VASP address	Destination not in label database	Show UNKNOWN + evidence + optional heuristic signals	"Our attribution is limited by label coverage; we expose that limitation instead of guessing."
False positive heuristic	Innocent wallet may look exchange-like	Show rule + confidence; require human review	"This is an investigative lead, not an accusation."
Mixer	Trace may become unreliable	Detect known mixer interaction and stop/flag	"We show exactly where reliable tracing ends."
Bridge / cross-chain	Single-chain graph ends at bridge boundary	Flag bridge and destination chain when observable	"Cross-chain continuation is a future capability."
API outage/rate limit	Live analysis may fail	Use cached demo data + backup case	"The live API is tested separately; the demo is reproducible."
Stale labels	VASP address mapping may change	Store source/date and refresh labels	"Attribution is source-backed and time-sensitive."
Private KYC needed	Blockchain cannot reveal account holder	Generate evidence for legal request	"Our tool identifies the VASP endpoint, not the customer identity."

§ 22 Post-Hackathon Roadmap		
Phase	Potential Enhancement	Dependency / Caution
Phase 1 — MVP (Current scope)	ETH + TRON, bounded tracing, VASP labels, interactive graph, PDF report, cached demo case.	Hackathon deliverable.
Phase 2	Larger label database, BSC, better heuristics, case storage, batch tracing, INR conversion.	Needs stronger data pipeline and testing.
Phase 3	Formal NCRP/I4C integration, secure institutional hosting, stronger audit trail, regional-language reports.	Requires official cooperation, security review and governance.
Phase 4	Advanced cross-chain analytics, ML entity clustering, live alerts, VASP collaboration.	Significant data, infrastructure and validation requirements.

Presentation rule: Only claim Phase 1 as BUILT. Present later phases as a roadmap, not as current capabilities.

§ 23

Six-Member Team Role Split

Member Role	Primary Responsibility	Deliverable
1. Backend / API Lead	Chain adapters, transaction retrieval, normalisation	Stable API endpoints + tests
2. Graph / Analytics Lead	NetworkX graph, traversal, path logic, heuristics	Trace engine + evidence structure
3. VASP / Data Lead	Label database, source verification, demo cases	Curated address dataset + source notes
4. Frontend Lead	React UI, Cytoscape graph, case input	Working investigator dashboard
5. Report / Integration Lead	ReportLab PDF, API/UI integration, caching	End-to-end report generation
6. Demo / Documentation Lead	Demo narrative, metrics, slides, Q&A,, backup assets	2-minute demo + judge defence

Team working principle: Do not let everyone build separate features independently. Define one internal data schema early so backend, graph, frontend and report generation all consume the same evidence object.

§ 24

Top 10 Judge Questions & Defensible Answers

Q
1

What is actually new here? Isn't this just a blockchain explorer?

A

A blockchain explorer exposes transaction data. Our prototype automates a focused investigation workflow: retrieve → trace → match supported VASP → explain evidence/confidence → generate report. The value is the end-to-end workflow, not any single component.

Q
2

Doesn't Chainalysis already do this?

A

Yes, professional platforms already provide powerful blockchain analytics. We are not claiming to replace them. Our student prototype focuses on a narrower, transparent workflow and a low-complexity investigator experience appropriate for the stated use case.

Q
3

Why would an investigator use this instead of Etherscan?

A

The value is reduced repetitive work: one input triggers bounded multi-hop tracing, entity matching, evidence presentation and report generation — instead of manual tab-by-tab navigation.

Q
4

How do you know the wallet belongs to a VASP?

A

Only when our address label evidence supports it. Exact database matches are stronger; behavioural heuristics are weaker and are shown separately with their rule and confidence.

Q
5

Can you identify the fraudster's real name?

A

No. Public blockchain data does not provide that identity. The system identifies a supported VASP endpoint and produces evidence that can support the appropriate legal/compliance request.

Q
6

Where does your data come from?

A

Public blockchain APIs and source-labelled public address data. The prototype does not depend on private government or exchange KYC data.

Q
7

What if the address is unknown?

A

We show UNKNOWN rather than guessing. We can still show the transaction path and transparent behavioural indicators, which support the investigator's next step.

Q
8

What about mixers and bridges?

A

We detect/flag known boundaries and stop or branch rather than fabricate a path. Full cross-chain and mixer analysis are outside the short MVP and are roadmap items.

**Q
9****How did you validate it?****A**

We use publicly verifiable cases and compare the reconstructed path/evidence with independent documentation. We also measure retrieval, graph, report and attribution performance on the test set and report actual observed figures.

**Q
1
0****What happens if your system is wrong?****A**

The system is an investigative aid. Every finding includes its source/confidence and requires human review before legal action. We do not claim automated prosecution.

§ 25

Quick-Start Implementation Checklist

Before Coding

- Freeze MVP: ETH + TRON first; BSC only if stable.
- Choose one real, public, verifiable demo case.
- Test the API response and save a local fixture.
- Prepare a small VASP label dataset with source/date.
- Agree on normalised transaction schema.
- Set maximum hop depth to 2–3.

Backend

- Address validation → chain selection → API adapter → normalised transactions → graph → traversal → VASP match → heuristics → evidence JSON.
- Add error handling and API rate-limit handling.
- Keep deterministic logic separate from optional LLM narration.

Frontend

- Single investigator input screen.
- Loading/error states.
- Graph + highlighted path.
- Node details and evidence panel.
- VASP confidence display.
- Report button.

Demo Hardening

- Cache demo data.
- Keep backup case and screenshots/video.
- Measure actual timings.
- Freeze new features before final rehearsal.

§ 26

Final Practical Verdict

Build this: The strongest student-level version is not a miniature Chainalysis. It is a reliable narrow system proving that a reported wallet can be converted into a traceable transaction graph, that a supported VASP endpoint can be identified when evidence exists, and that the investigator receives a structured next-step report.

Why the Project Is Feasible

- The PS is specific enough to define a clear MVP.
- Public blockchain data can support the core demonstration.
- Graph traversal and visualisation are mature software components.
- A small source-labelled VASP dataset is enough for a controlled demo.
- The end-to-end flow is visually demonstrable.
- The project can be kept software-first with a bounded 3–5 day scope.

Biggest Weaknesses to Acknowledge

- VASP label coverage is limited compared with commercial intelligence providers.
- Unknown wallets cannot be confidently attributed just from behaviour.
- Cross-chain, mixer and privacy-enhancing mechanisms can break or complicate the trace.
- External API availability and rate limits can threaten a live demo.
- Legal admissibility and identity attribution are outside the prototype.

Winning principle: Demonstrate a working narrow system with honest evidence rather than a huge system with unverified claims. Your strongest story is: reported wallet → automated fund-flow trace → supported VASP endpoint → evidence/confidence → actionable report.

Final boundary: CryptoTrace I4C generates investigative intelligence. It does NOT automatically freeze funds, reveal KYC identity, prosecute anyone, or guarantee that an unknown address belongs to a specific person or exchange.

§ 27 Team Glossary — Say It Simply

Term	Meaning
Address	Crypto identifier used as a transaction endpoint.
Asset / token	The crypto being transferred, e.g. USDT.
Hop	One wallet-to-wallet step in a traced path.
Node	A wallet/entity represented in the graph.
Edge	A transaction connecting two nodes.
VASP	Virtual Asset Service Provider; in this context, a crypto service/exchange.
Attribution	Associating an address with a known entity based on evidence.
Heuristic	A rule that suggests a pattern; not proof.
Confidence	How strong the evidence for an attribution is.
Label database	List mapping known addresses to entities, with source information.
N-hop tracing	Explore the graph only up to a fixed number of steps.
Entity attribution	Identifying a service/entity, not a person.
KYC	Customer identity information held by a service.
Mixer	Service designed to make fund provenance harder to follow.
Bridge	Mechanism that moves assets between blockchain networks.
Evidence hash	Transaction identifier that lets someone verify the underlying transfer.

§ 28

One-Page Team Cheat Sheet

Problem	Victim reports a suspect crypto wallet. Investigators need to follow the money and identify the nearest
Solution	One address in → public transaction data → bounded graph trace → VASP label match → confidence
Innovation	Not 'AI.' The differentiation is a focused, transparent investigator workflow that compresses repetitive l
What we CAN prove	Transaction path, transaction hashes, known-address attribution, observable patterns, processing time
What we CANNOT claim	Real person identity • Guaranteed unknown-wallet attribution • Complete cross-chain tracing • Mixer d
MVP scope	ETH + TRON, 2-3 hops, small source-labelled VASP dataset, interactive graph, evidence panel, PDF

The 10-Second Explanation to a Judge

"We start with the wallet a victim reports. Instead of making an investigator trace every transaction manually, our system automatically builds the fund-flow graph, finds a supported VASP endpoint when the evidence allows it, shows why we believe that result, and generates the investigation evidence in one workflow."

Verification Notes

Before the team uses any claim in the final PPT or judge answer, verify:

- Current SIH/I4C wording and any official evaluation requirements.
- Current API endpoints, authentication requirements and rate limits.
- Current public VASP/exchange address labels.
- Any numerical claim about Indian cybercrime volumes or crypto-fraud patterns.
- Any claim about a specific company's current pricing, chain coverage or product features.
- The exact public transaction path used in the demo case.
- Any statement about legal procedure, freezing, evidence admissibility or exchange cooperation.

Final reminder: If a judge asks something you cannot prove, say so. A transparent limitation is stronger than an invented answer.